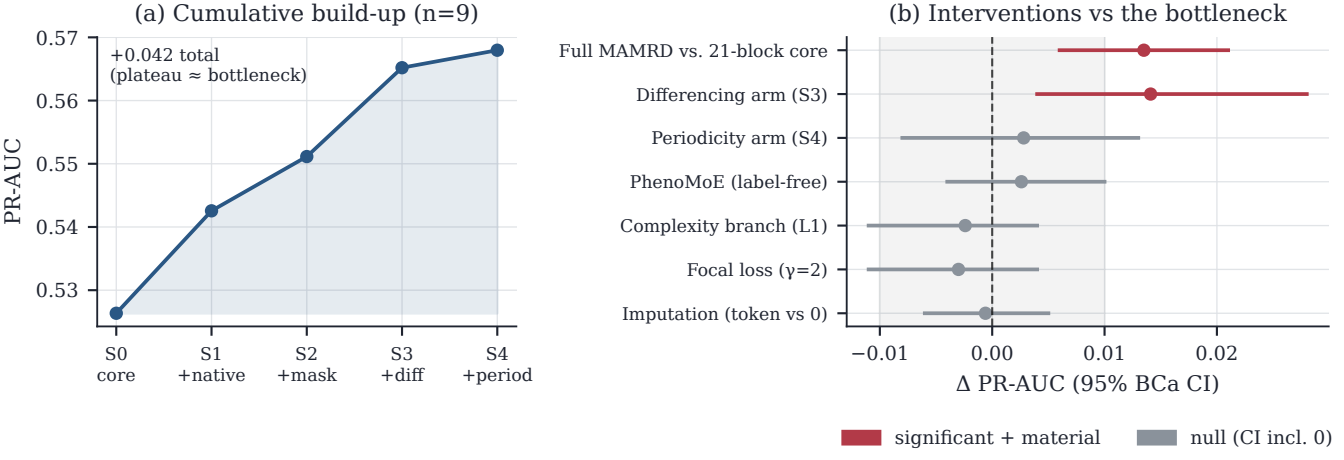
(a) Cumulative build-up (n=9)
PR-AUC
0.57
0.56
0.55
0.54
0.53
+0.042 total
(plateau ≈ bottleneck)
S0
core
S1
+native
S2
+mask
S3
+diff
S4
+period
(b) Interventions vs the bottleneck
Full MAMRD vs. 21-block core
Differencing arm (S3)
Periodicity arm (S4)
PhenoMoE (label-free)
Complexity branch (L1)
Focal loss (γ=2)
Imputation (token vs 0)
−0.01
0.00
0.01
0.02
Δ PR-AUC (95% BCa CI)
significant + material
null (CI incl. 0)